



Interactive Sports Analytics with Power BI: A Data Visualization Project

Prepared as part of Data Analytics and Visualization Project

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Chapter 1

Introduction & Objective

This Power BI project aims to transform raw sports data into actionable intelligence. We focus on comprehensive datasets encompassing player statistics, match outcomes, and various performance metrics. Our core objective is to develop an interactive dashboard that offers dynamic visualizations and deep insights into evolving sports performance trends.

The dashboard will enable sports analysts and coaching staff to identify key patterns, evaluate player and team efficiencies, and make data-driven decisions to enhance competitive strategies. We are moving beyond static reports to provide a living, breathing analytical tool.

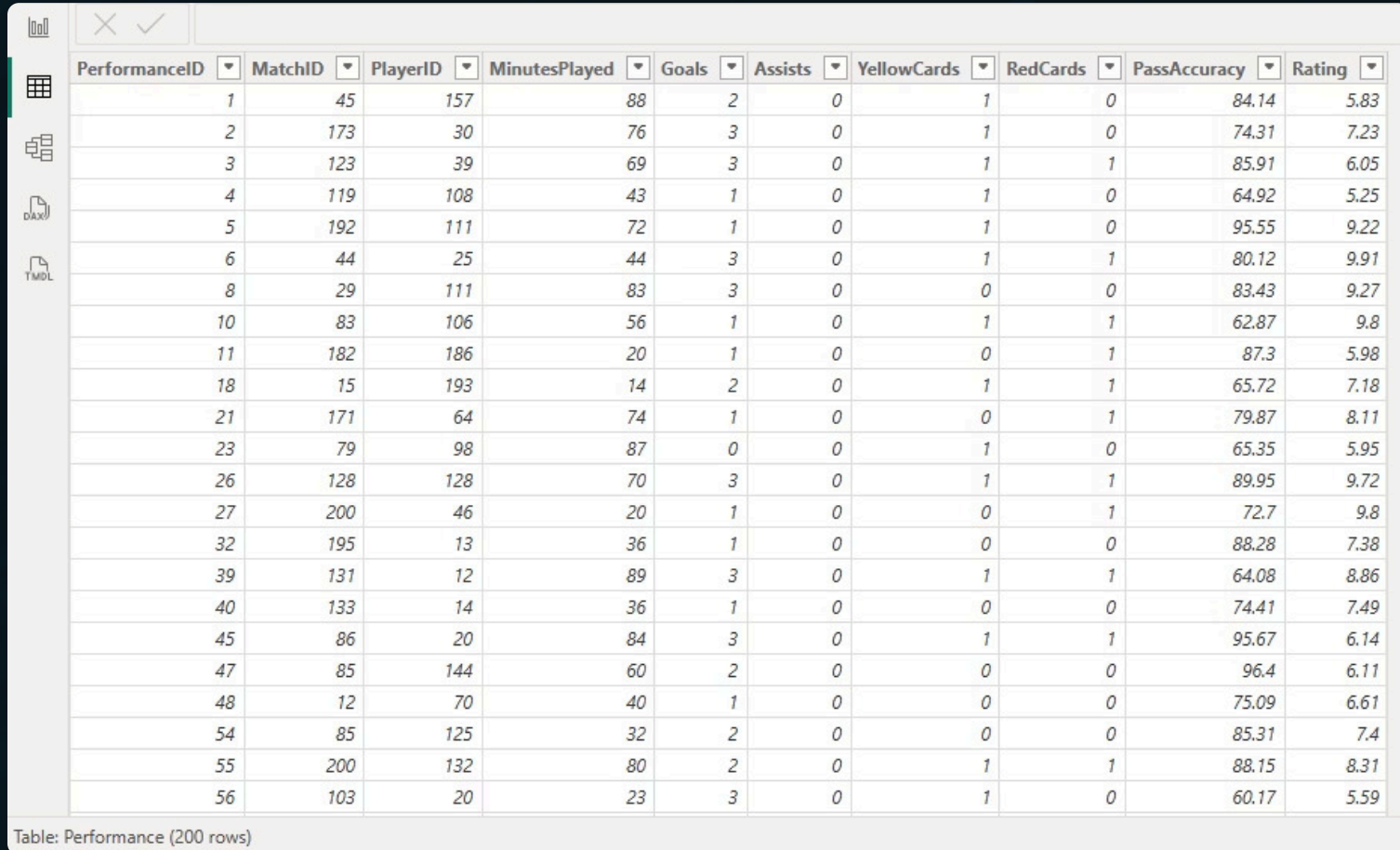


Raw data of Matches dataset

MatchID	Date	HomeClub	AwayClub	HomeGoals	AwayGoals	Stadium	Referee	Attendance	City	Country	Latitude	Longitude
2	02 January 2023	ClubD	ClubA	4	1	Stadium1	Ref_2	44084	London	England	51.5074	-0.1278
3	03 January 2023	ClubC	ClubD	4	3	Stadium1	Ref_3	55105	Munich	Germany	48.1351	11.582
6	06 January 2023	ClubB	ClubD	3	3	Stadium4	Ref_6	17227	Paris	France	48.8566	2.3522
7	07 January 2023	ClubA	ClubB	0	2	Stadium2	Ref_7	41910	London	England	51.5074	-0.1278
8	08 January 2023	ClubC	ClubB	2	2	Stadium2	Ref_8	46041	Munich	Germany	48.1351	11.582
9	09 January 2023	ClubA	ClubC	2	3	Stadium4	Ref_9	33384	London	England	51.5074	-0.1278
12	12 January 2023	ClubC	ClubB	2	1	Stadium1	Ref_12	26035	Madrid	Spain	40.4168	-3.7038
13	13 January 2023	ClubD	ClubB	4	2	Stadium1	Ref_13	13040	Paris	France	48.8566	2.3522
14	14 January 2023	ClubB	ClubC	0	0	Stadium1	Ref_14	52422	Munich	Germany	48.1351	11.582
15	15 January 2023	ClubD	ClubC	4	3	Stadium1	Ref_15	36160	Paris	France	48.8566	2.3522
18	18 January 2023	ClubA	ClubB	2	1	Stadium1	Ref_18	39344	Madrid	Spain	40.4168	-3.7038
19	19 January 2023	ClubC	ClubD	3	3	Stadium1	Ref_19	27014	London	England	51.5074	-0.1278
20	20 January 2023	ClubB	ClubD	4	3	Stadium1	Ref_20	24996	Munich	Germany	48.1351	11.582
21	21 January 2023	ClubA	ClubC	2	4	Stadium3	Ref_21	59104	Madrid	Spain	40.4168	-3.7038
22	22 January 2023	ClubA	ClubC	4	0	Stadium1	Ref_22	51621	Munich	Germany	48.1351	11.582
23	23 January 2023	ClubB	ClubC	1	4	Stadium3	Ref_23	21695	Paris	France	48.8566	2.3522
24	24 January 2023	ClubA	ClubB	0	2	Stadium4	Ref_24	12852	Munich	Germany	48.1351	11.582
26	26 January 2023	ClubB	ClubA	1	2	Stadium1	Ref_26	37042	Madrid	Spain	40.4168	-3.7038
28	28 January 2023	ClubB	ClubA	2	4	Stadium2	Ref_28	50390	Paris	France	48.8566	2.3522
29	29 January 2023	ClubD	ClubA	4	4	Stadium4	Ref_29	33711	Paris	France	48.8566	2.3522
31	31 January 2023	ClubD	ClubA	3	2	Stadium3	Ref_31	19160	London	England	51.5074	-0.1278
32	01 February 2023	ClubB	ClubD	1	3	Stadium2	Ref_32	37213	Munich	Germany	48.1351	11.582
33	02 February 2023	ClubC	ClubA	4	1	Stadium2	Ref_33	49420	London	England	51.5074	-0.1278

Table: Matches (2) (149 rows)

Raw data of Performances dataset



PerformanceID	MatchID	PlayerID	MinutesPlayed	Goals	Assists	YellowCards	RedCards	PassAccuracy	Rating
1	45	157	88	2	0	1	0	84.14	5.83
2	173	30	76	3	0	1	0	74.31	7.23
3	123	39	69	3	0	1	1	85.91	6.05
4	119	108	43	1	0	1	0	64.92	5.25
5	192	111	72	1	0	1	0	95.55	9.22
6	44	25	44	3	0	1	1	80.12	9.91
8	29	111	83	3	0	0	0	83.43	9.27
10	83	106	56	1	0	1	1	62.87	9.8
11	182	186	20	1	0	0	1	87.3	5.98
18	15	193	14	2	0	1	1	65.72	7.18
21	171	64	74	1	0	0	1	79.87	8.11
23	79	98	87	0	0	1	0	65.35	5.95
26	128	128	70	3	0	1	1	89.95	9.72
27	200	46	20	1	0	0	1	72.7	9.8
32	195	13	36	1	0	0	0	88.28	7.38
39	131	12	89	3	0	1	1	64.08	8.86
40	133	14	36	1	0	0	0	74.41	7.49
45	86	20	84	3	0	1	1	95.67	6.14
47	85	144	60	2	0	0	0	96.4	6.11
48	12	70	40	1	0	0	0	75.09	6.61
54	85	125	32	2	0	0	0	85.31	7.4
55	200	132	80	2	0	1	1	88.15	8.31
56	103	20	23	3	0	1	0	60.17	5.59

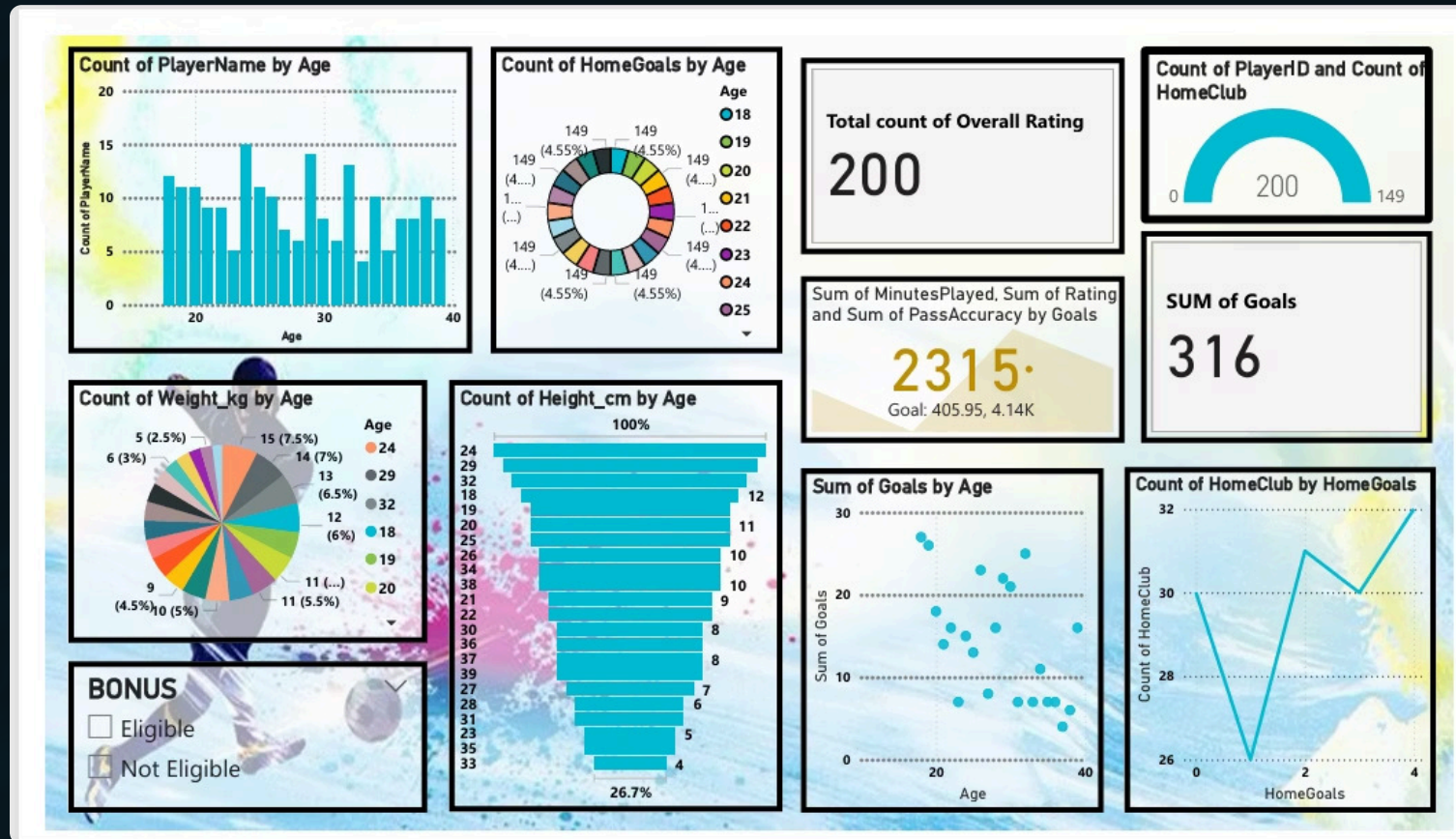
Table: Performance (200 rows)

Raw data of Players dataset

PlayerID	PlayerName	Age	Position	Height_cm	Weight_kg	Nationality	Club	OverallRating	BONUS
3	Player_3	32	Goalkeeper	183	73	France	ClubB	64	Not Eligible
8	Player_8	36	Goalkeeper	186	88	Spain	ClubD	59	Not Eligible
10	Player_10	28	Goalkeeper	185	77	Spain	ClubB	90	Eligible
16	Player_16	38	Goalkeeper	188	66	USA	ClubB	51	Not Eligible
17	Player_17	19	Goalkeeper	185	72	Spain	ClubD	56	Not Eligible
20	Player_20	19	Goalkeeper	183	86	France	ClubA	83	Eligible
23	Player_23	29	Goalkeeper	195	88	Brazil	ClubB	92	Eligible
26	Player_26	34	Goalkeeper	167	78	France	ClubA	74	Not Eligible
27	Player_27	27	Goalkeeper	175	67	France	ClubD	56	Not Eligible
29	Player_29	32	Goalkeeper	171	81	Germany	ClubA	73	Not Eligible
35	Player_35	22	Goalkeeper	191	65	USA	ClubA	75	Not Eligible
36	Player_36	36	Goalkeeper	182	90	France	ClubA	77	Not Eligible
38	Player_38	38	Goalkeeper	184	98	Germany	ClubC	70	Not Eligible
41	Player_41	35	Goalkeeper	165	85	Brazil	ClubB	66	Not Eligible
42	Player_42	21	Goalkeeper	174	78	USA	ClubB	69	Not Eligible
45	Player_45	26	Goalkeeper	167	72	France	ClubA	69	Not Eligible
48	Player_48	37	Goalkeeper	163	87	Brazil	ClubB	89	Eligible
58	Player_58	35	Goalkeeper	176	65	USA	ClubA	79	Not Eligible
59	Player_59	25	Goalkeeper	178	91	USA	ClubC	78	Not Eligible
61	Player_61	19	Goalkeeper	185	75	France	ClubC	94	Eligible
63	Player_63	39	Goalkeeper	185	70	England	ClubA	79	Not Eligible
64	Player_64	27	Goalkeeper	182	96	England	ClubA	65	Not Eligible
68	Player_68	29	Goalkeeper	160	79	Brazil	ClubB	50	Not Eligible

Table: Total count (200 rows)

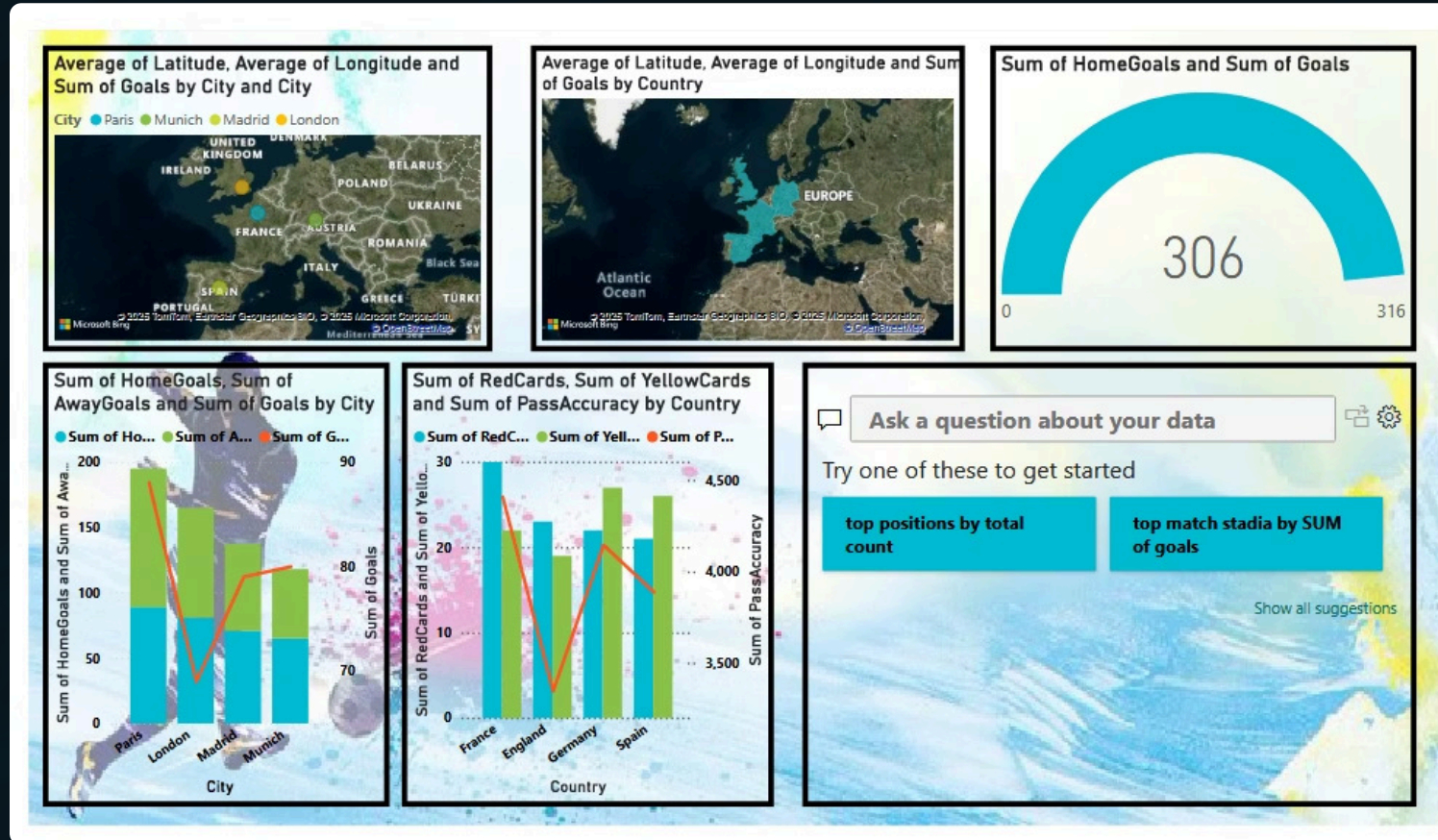
Dashboard Page 1



Dashboard Page 2



Dashboard Page 3



Key Results & Insights

Player Performance Trends

Deep dive into individual player statistics across multiple matches, revealing consistent performers and areas for improvement.

Team Comparison

Comparative analysis of teams based on critical metrics like scoring efficiency, defensive prowess, and historical win/loss ratios.

Geographic Analysis

Visualize the distribution of matches and venues, identifying regional strengths and strategic logistical insights.

Top Performers Identification

Automated identification of top-performing players and teams through advanced ranking algorithms.



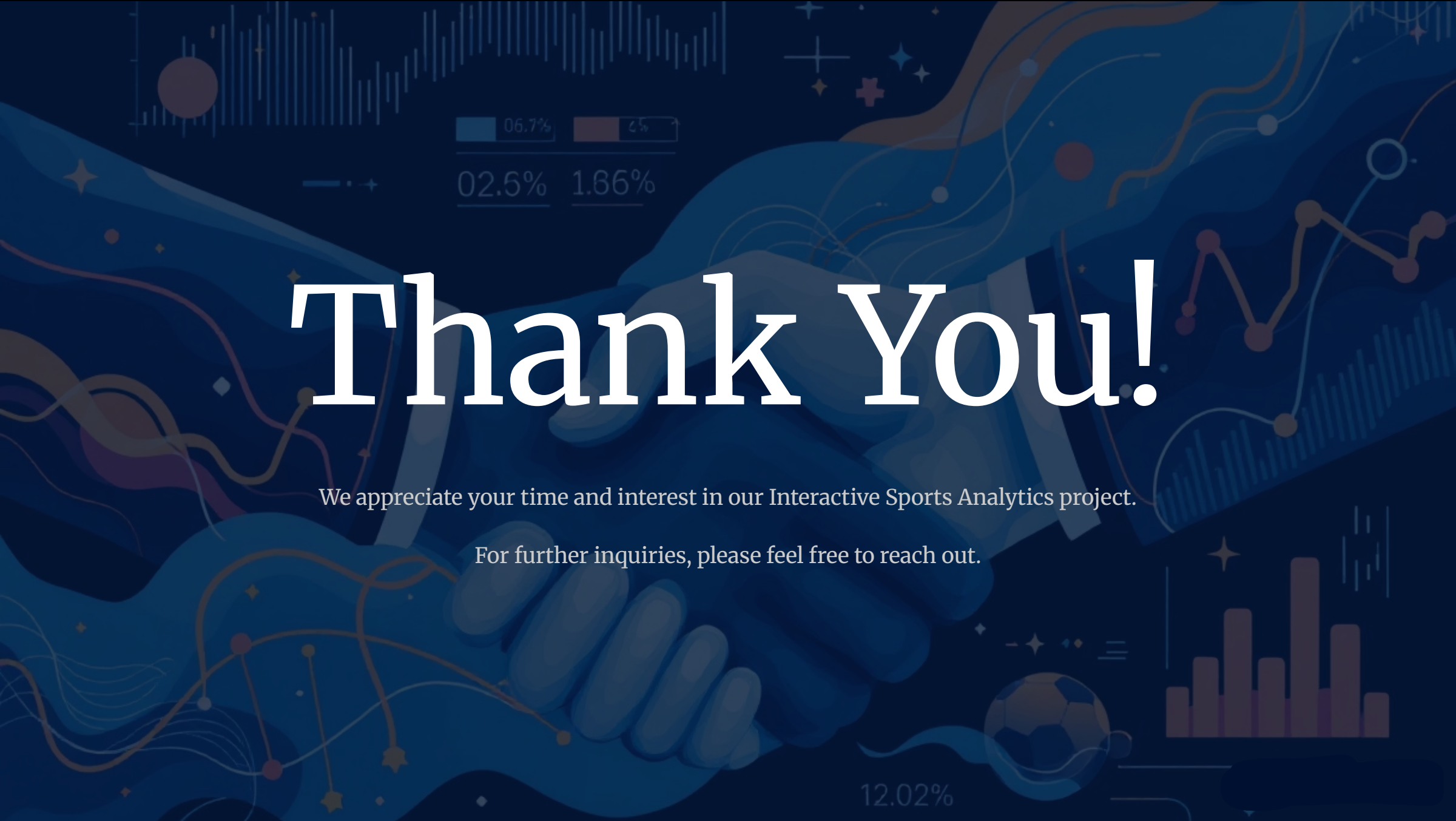
The dashboards are meticulously designed with a Dark Modern Blue theme for a professional and contemporary aesthetic, ensuring optimal data readability.

Conclusion

This project underscores the transformative power of data visualization in the realm of sports analytics. By leveraging the robust capabilities of Power BI, we have successfully engineered an interactive and visually compelling dashboard. This dashboard dramatically simplifies the process of analyzing complex sports datasets, providing an intuitive interface for deep insights.

It serves as a testament to how advanced analytical tools can empower sports organizations to unlock hidden patterns, optimize performance, and ultimately gain a competitive edge. The insights gleaned from this dashboard can drive strategic decisions, player development, and overall team success.



The background is a dark blue gradient featuring a central illustration of two hands shaking. Surrounding this are various sports analytics-themed elements: a line graph at the top left, a bar chart at the top right, a soccer ball at the bottom right, and several smaller charts and percentages scattered throughout. The text 'Thank You!' is prominently displayed in the center in a large, white, serif font.

Thank You!

We appreciate your time and interest in our Interactive Sports Analytics project.

For further inquiries, please feel free to reach out.